

GED Math: Number Sense & Problem Solving

20 Practice Questions with Full Solutions

November 6, 2025

Instructions

Show your work. Calculators are allowed unless your instructor says otherwise.

Practice Questions

1. Convert 0.625 to a fraction in simplest form.
2. Simplify the fraction $\frac{84}{126}$.
3. Round 7,495 to the nearest hundred.
4. Round 0.01986 to the nearest thousandth.
5. Classify each number as integer, rational, or irrational (choose all that apply): -12 , $\sqrt{2}$, $0.\bar{3}$, π .
6. Estimate $398 + 603$ using rounding.
7. Estimate 51×19 using compatible numbers.
8. Approximate $5,982 \div 29$ using friendly numbers.
9. Write the ratio of 18 students to 12 students in simplest form.
10. Solve the proportion $\frac{x}{15} = \frac{4}{5}$.
11. Find 17% of 260.
12. The price of a jacket increases from \$48 to \$60. What is the percent increase?
13. A pair of shoes costs \$85 and is discounted 30%. After the discount, 8% sales tax is added. What is the total price?
14. A recipe uses 3.6 liters of lemonade to serve 9 people. How many liters are needed to serve 15 people (same rate)?
15. Solve for x : $3(x - 4) + 5 = 20$.
16. A cyclist travels at an average speed of 12 mph for 1.75 hours. How far did they travel?

17. Pump A can fill a tank in 8 hours, and Pump B can fill it in 12 hours. Working together, how long will it take to fill the tank?
18. A solution contains 25% salt by volume. How much water must be added to 200 mL of this solution to obtain a 10% salt solution? (Assume volumes are additive.)
19. You work 38 hours at \$14 per hour. After a 12% deduction, what is your take-home pay?
20. A muffin recipe needs 2.5 cups of flour to make 16 muffins. How many cups are needed to make 24 muffins (same recipe)?

Full Solutions

1. $0.625 = \frac{625}{1000} = \frac{125}{200} = \frac{25}{40} = \boxed{\frac{5}{8}}.$
2. $\gcd(84, 126) = 42. \frac{84}{126} = \frac{84 \div 42}{126 \div 42} = \boxed{\frac{2}{3}}.$
3. Nearest hundred: the tens digit is 9 (round up). $\boxed{7,500}.$
4. Thousandth place is the third decimal digit. 0.01986 has 8 in the next place, so round up to $\boxed{0.020}.$
5. -12 : integer and rational. $\sqrt{2}$: irrational. $0.\bar{3} = \frac{1}{3}$: rational. π : irrational.
 $\boxed{\text{Ans: } -12 \in \mathbb{Z} \cap \mathbb{Q}; \sqrt{2}, \pi \text{ irrational; } 0.\bar{3} \in \mathbb{Q}.}$
6. $398 \approx 400, 603 \approx 600 \Rightarrow 400 + 600 = \boxed{1000}.$ (Exact sum = 1001.)
7. Compatible numbers: $51 \approx 50, 19 \approx 20 \Rightarrow 50 \times 20 = \boxed{1000}.$ (Exact product = 969.)
8. Friendly numbers: $5,982 \approx 6,000, 29 \approx 30 \Rightarrow 6,000 \div 30 = \boxed{200}.$ (Exact quotient = 206.275...)
9. $18 : 12 = \frac{18}{12} = \frac{3}{2} \Rightarrow \boxed{3 : 2}.$
10. Cross-multiply: $5x = 4 \cdot 15 = 60 \Rightarrow x = \boxed{12}.$
11. $0.17 \times 260 = \frac{17}{100} \times 260 = \frac{17 \times 260}{100} = \frac{4420}{100} = \boxed{44.2}.$
12. Change = $60 - 48 = 12$. Percent increase = $\frac{12}{48} = 0.25 = \boxed{25\%}.$
13. Discounted price = $85(1 - 0.30) = 85(0.70) = \59.50 . Tax = $0.08 \times 59.50 = \$4.76$. Total = $59.50 + 4.76 = \boxed{\$64.26}.$
14. Rate = $\frac{3.6}{9} = 0.4$ L/person. For 15 people: $0.4 \times 15 = \boxed{6 \text{ L}}.$
15. $3(x - 4) + 5 = 20 \Rightarrow 3x - 12 + 5 = 20 \Rightarrow 3x - 7 = 20 \Rightarrow 3x = 27 \Rightarrow x = \boxed{9}.$
16. $d = rt = 12 \times 1.75 = \boxed{21 \text{ miles}}.$
17. Rates add: $\frac{1}{8} + \frac{1}{12} = \frac{3}{24} + \frac{2}{24} = \frac{5}{24}$ tank/hour. Time = $\frac{1}{\frac{5}{24}} = \boxed{\frac{24}{5} = 4.8 \text{ hours}}.$
18. Salt in original: $0.25 \times 200 = 50$ mL. Let final volume be V . Want 10%: $0.10V = 50 \Rightarrow V = 500$ mL. Water added = $500 - 200 = \boxed{300 \text{ mL}}.$
19. Gross pay = $38 \times 14 = \$532$. Deduction = $0.12 \times 532 = \$63.84$. Take-home = $532 - 63.84 = \boxed{\$468.16}.$
20. Scale factor = $\frac{24}{16} = 1.5$. Flour = $2.5 \times 1.5 = 3.75 = \boxed{3\frac{3}{4} \text{ cups}}.$